HomologyRing: A Python Tool for Analyzing Intra- and Inter-Chain Contact Specificity in Protein Families.

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Abstract

Contacts formed by peptides are principally responsible for stabilizing protein structure and determining protein function. AlphaFold 2, developed by DeepMind has proven capable of predicting protein structure with unprecedented accuracy. With predictions available for the majority of known proteins, AlphaFold has proven transformative for bioinformatics, resulting in an abundance of high-quality structural data. RING is a software that deterministically predicts non-covalent interactions in such structure files, providing high accuracy contact information with a wide verity of uses including: aiding in human understanding of protein structure or function, and training machine learning models. HomologyRing is a Python package that combines the functionality of RING with a homology search, enabling the analysis of contact conservation and variance across many of evolutionarily related proteins. Starting from a query structure or sequence, HomologyRing uses BLAST to perform a homology search against either UniProt or Protein Data Bank (PDB) databases, and subsequently using RING to collect contact information for corresponding AlphaFold or PDB structures. We demonstrate the utility of the resulting homology-enriched contact information for characterizing how preservation and variance of contacts in homologs contributes to protein structure, function, and partner binding. By mapping contacts to corresponding residues in a multiple sequence alignment (MSA), HomologyRing compiles and visualizes detailed information on intra- and inter-chain contacts, with wide potential applications, including: study of DNA, ligand, and partner binding specificity.

Contents

Abbreviations and Terminology

Abbreviation	Definition
MSA	Multiple Sequence Alignment
RIN	Residue Interaction Network
hRIN	Homology Residue Interaction Network
PDB	Protein Data Bank
RING	Residue Interaction Network Generator [3]

We interchangeably use the following terms depending on context: *interaction*, *non-covalent interaction*, and *edge*.
Types of non-covalent interactions are referred to by RING identifiers. A complete list of which can be found in Appendix B

1 Introduction

This section assumes familiarity with central biological dogma, as well as knowledge of basic concepts in Bioinformatics. Extensive background is provided in Appendix A that aims to accomidate readers of various backgrounds. The topics particularly relevant to this work are explained in the following sections.

1.1 Residue Interaction Networks

At the very basic level, protein chains are polymers of amino acid residues bound together by covalent bonds. As a chain folds, it is *non-covalent* interactions between residues that are principally responsible for the secondary and tertiary structure (conformation) of the protein. These non-covalent interactions could be a verity of different forces and are the result of the different chemical and physical properties of the 20 different base amino acids and their arrangement in the peptide sequence that constitutes the protein chain. Examples of interactions commonly observed witnin proteins include Van Der Walls forces and hydrogen bonding. Other interactions are observed less frequently and exibihit high specificity with respect to amino acid participants, like Ionic bonds or $\pi - \pi$ interactions and several others. Furthermore, the structure of a protein is intrensically tied to it's function within its biological context. With this in mind, there's a lot of information that can be gathered from knowledge of these non-covalent interactions. Researchers may very easily interpret the structural or functional significance of an interaction, but methods concerning non-covalent interactions can be applied to a problem long troubling structural bioinformatics: The amount of available structural information has been growing at a near exponential rate. Both experimental structures available through the PDB and high accuracy predictions given by AlphaFold2 are commonly available in PDB or mmCIF file formats. These files are large—containing 3D atom data, chemical information, and a host of metadata. Parsing, and algorithmically extracting functionally relevant information algorithmically proves to be an arduous task in many contexts. One approach to addressing this problem has garnered much attention: The construction of Residue Interaction Networks (RIN) use information of non-covalent interactions to create a simplified representation of a protein structure. There are many variants of the idea. However, they are generally defined for a protein structure to be a mathematical graph, or network, denoted $G = (N, E)$ where N is a set of nodes identifying amino acid residues or various compounds in a structure and E is taken the set of non-covalent interactions observed in the given structure. An example of such a network is shown in figure 1. RINs can be seen to provide a simplified means of capturing the topology of a protein structure, and has seen notable attention for its various applications: Binding, protein engineering... Training data of machine learning methods like graph neural networks (GNN). Historical methods like Convolutional neural networks (CNN) on distance matricies discard physicochemical information and may be considerable larger than analogous representation of the same structure with a RIN.

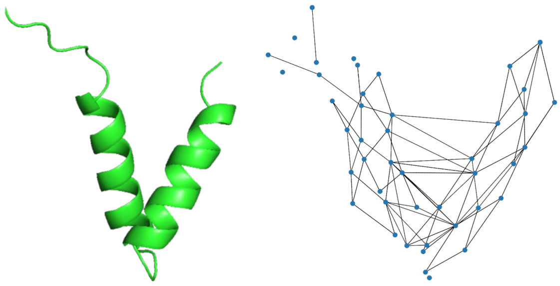

Figure 1: Example of a basic RIN for generated for a protein chain.

If you are going to weight edges with prob for ensembles, it should go here.

1.2 Residue Interaction Network Generator - RING

There are many available tools capable of generating RINs, each with a slightly different interpretation of the object. In this work, such networks are created using Residue Interaction Network Generator - RING. Notable for its ability to handle all ligands and modified residues in the PDB, it is regarded for accurately and deterministically predicting non-covalent interactions within structure files. It is worth noting that the output of RING is a multi-Graph. Differing from the typical mathematical notion of a graph, it permits multiple edges between a given pair of nodes (see Figure 2). These different edges correspond to the different non-covalent interactions predicted by RING. The list of interactions predicted by RING comprising the possible edge types in the resulting RINs are described in Appendix B. Furthermore, we will refer to the various interaction types by their internal RING identifiers throughout this text for the sake of brevity and clarity (e.g. `HBOND` for hydrogen bonding and `IONIC` for salt bridges).

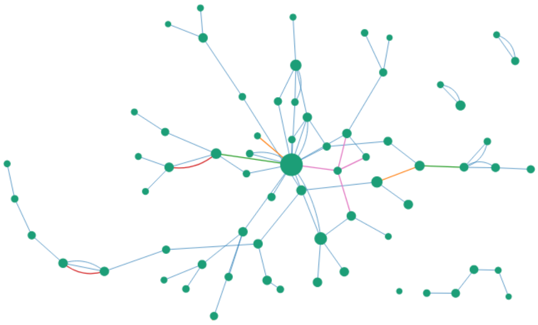

Figure 2: Example of a RIN multi-graph visualization created with RING. As typical with RINs, nodes correspond to amino acid residues in a given protein structure. Edges correspond are non-peptide interactions or bonds between residues. Where edges of many interactions are types are permitted between a pair of nodes.

2 Pipeline

The tool is implemented as a python package responsible for running the pipeline and conducting supporting analysis. The execution of the pipeline to construct a hRIN represents the primary responsibility of the python object and must be ran after object initialization to perform any useful subsequent action on a hRIN.

An overview of the pipeline steps are as follows:

1. The pipeline is initiated for a query protein chain identified in a CIF file or a list representing a family of homolog chains is provided by the user.
2. For a query chain, AA sequence is used as a query for a BLAST homolgy search of either the AlphaFold or PDB database.
3. Retrieve Structures for homolog proteins.
4. Multiply align sequences of homolog chains.
5. Predict non-covalent interactions for each structure with RING.
6. Synthesize homolog RINs with alignment information to create homology-aware hRIN.

2.1 Initiation

The construction of the hRIN is initiated on either a user-chosen query chain or a user supplied family of structure. In the former situation, the query chain is taken to be a representative example of the protein family - the subject of study that will form the basis of the homolog search and construction of the Interaction network. To begin, The user supplies a Crystallographic Information File (CIF) that contains the physical information about the query chain. From this file the query polypeptide sequence is extracted and processed to address potential non-standard or unknown residues present in the sequence. Specifically, CIF files will contain a *canonical sequence* for the chain that is the result of some rudimentary processing, simplifying non-standard residues. In the ladder case, the pipeline supports analysis of a user defined of protein chains. In this case, the user gives a list of structures which are known a priori to contain homolog chains, forgoing the need for a homolgy search. The user must additionally supply a list identifying which chains withing the structures are the homologs to be considered for the construction of the hRIN (by their internal CIF chain identifiers: label_asym_id) in the form of a tsv file containing the name of the structure and the chain ID.

2.2 Homology Search

With the query sequence extracted, it is used to perform a homology search with protein-BLAST. The user has the choice to perform this search over either the database of UniProt predictions or the Protein Databank (PDB). These database differ in their distribution of contained sequences - with the protein databanks around XXXX protein sequences over XXXX structures representing a notably more biased population than uniprots XXXX protein sequences. However, the next step is to automatically download structures for the returned homolog sequences. Metadata describing the map of sequences to their homolog chains in corresponding structures as well as matched regions from the BLAST results are maintained and used for the later construction of the final hRIN.

2.3 Structure Retrieval

Due to unreliable experimental structure availability for all of uniprot, choosing to perform the homolog search over the larger database will necessitate that homolog structures be retrieved from the database of AlphaFold2 predictions, which has the notable limitation of contain structures for only a single protein chain. Where searching for sequences over the PDB will result in the pipeline downloading the corresponding experimental structures. These may contain comples of protein chains, other polymers (such as nucleotides) and Ligands. This permits many potential useful analyses.

Once homolog structures are collected, some basic filtering is performed to ensure CIF structures are not malformed, as is sometimes the case in the PDB, preventing the running of RING due to missing secondary structure information, for example. The peptide sequences of each homolog chain are extracted from the structure. This ensures parody with information contained in the hRIN and the component structures used for its construction, thus avoiding issues caused by out of data BLAST databases and updated PDB/UniProt entries can causing discrepancies. Discussion of the different structure databases can be seen in Section 3.

In later sections, we regularly refer to a family of structures wich to take to mean the set of structures returned by this process. finish

contains a what we regularly refer to as a *homolog chain* that corresponds to the a sequences identified during the homology search due to its high sequence similarity to the query sequence. In addition to the *homolog chain*, structures in the family, if retrieved from the PDB, are commonly complexes of many entities. It is common that a complex may contain many homolog chains, but we will refer to entities that are not homologous to the query sequence as *non-homolog entities*. These entities are further categorized in Subsection 4.2.

2.4 Sequence Alignement

The full chain sequences resulting from the operation (not just the sub-sequences returned by BLAST) are aligned into a multiple sequence alignment (MSA) using clustalΩ. This MSA is later used for the identification of nodes within the hRIN.

Multiple Sequence Alignments function to identify families of corresponding residues in many sequences. They assume

2.5 RING

Running RING on each homolog structure results in a family of RINs. However, sequence variations across the homolog chains mean that a homology-aware method for synthesizing their information is necessary for the construction of a hRIN. To do this nodes are either defined to be a family of homologous residues or, or a set of non-homolog entities defined across the set of structures. This requires a system of maps relating residues identified in structures as well as indices in BLAST and RING output to columns in the MSA. Non-homolog chain entities are mapped to nodes in the hRIN through a parallel process. The details of node definition are described in Section 4 and Subsection 4.2.

2.6 hRIN Synthesis

The final step in the core of the Ring Homology pipeline is the synthesis of the hRIN from previously collected information. The most import aspects of this process include merging homology information from the BLAST search, alignment information from ClustalΩ, and interaction information from RING.

3 Homology Database

3.1 AlphaFold

AlphaFold is model for predicting the 3D structure of a protein chain given its amino-acid sequence, a challenging and long standing problem in bioinformatatics. AlphaFold2's predictions are renown for their high accuracy, and have proven transformative for the field [4]. AlphaFold2 has the notable restriction that predictions are formed for only a single protein chain, not entire complexes of multiple protein chains or ligands - non-polymer structures with which a protein chain is known to bind or interact. They tend to be of high functional significance, and the subject of many analyses. Ligands are commonly resolved in experimental methods catalogued in the PDB like MNR or X-ray crystallography. AlphaFold3 predictions include protein complexes and protein-peptide interactions, but have yet to be made widely available for all UniProt sequences via a database.

3.2 Protein Data Bank (PDB)

4 Homology Residue Interaction Network - hRIN

We use the term Homology Interaction Network (hRIN) to refer to homology enriched RINs. It should be noted that nodes witin the hRIN do not identify individual residues as they do in typical RINs. Rather, they generally correspond to families of residues as identified by columns in the MSA. In other works this similar concepts of using an MSA to synthasise a set of RINs are called Key Interaction Networks (KINs) [9].

4.1 Contact Conservation

On of the key aspects of homology information encoded to hRINs are edges weighted with the probability or conservation score. This value captures the propensity of an interaction represented by an edge being present in any given member chain of the protein family. This required developing different notions of *contact conservation* that can be interchanged depending on the needs of the user. The first, more basic, definition simply normalizes the number of observed interactions of a given type for a given pair of nodes by the number of homolog structures in the family. That is, for residue nodes $u, v \in N$ and interaction type $\xi \in \{\texttt{VDW}, \texttt{HBOND}, ...\}$ [1] (see appendix B) , we define:

$$p_\xi(u, v) = \frac{1}{|H|} \sum_{x \in H} I_\xi(u, v, x)$$

where N is the number of homolog structures in the family and I_ξ is simply the indicator function: returning 1 if nodes u and v are connected by an edge of type ξ, else 0. This does well to normalize interaction counts to probabilities, but has the effect of diminishing some interactions conservation score when it may, in fact, not even be possible in every structure. This is because some nodes may represent families of families of residues given by columns of the MSA with low occupancy, meaning they have gaps or no corresponding residue for some structures. Given the variation seen in the PDB for both indels in sequences, and entities present in complexes, it may be undesirable to penalize such situations for some analyses. To address this, we provide an alternative formulation of conservation:

$$p_\xi(u, v) = \sum_{x \in H} \frac{I_\xi(u, v, x)}{I_{\text{non-gap}}(u, x) \cdot I_{\text{non-gap}}(v, x)}$$

[1] The HomologyRing tool supports calculating conservation socres for multiple interaction types at once to support analysis requiring queries selecting multiple interaction types of interest. It only requires that conservation scores also be normalized by the number of selected interactions.

4.2 Inter-Chain & Ligand Node Identification

Identifying nodes by families of residues given by columns of the MSA works well for nodes associated with residues in the query structure. However, many structures in the PDB contain a complex of multiple *entities* and many potential analyses performed with RIN concern the interaction of a complex of multiple protein chains and ligands. In this context *entity* refers to a structural unit with in structure files. They may be polymer chains (protein or nucleotide) or ligands. Ligands are small non-polymer molecules, ions or other chemical entities that bind or otherwise interacting with sites of a peptide structure. These can be endogenous molecules like co-factors or signaling molecules. We make the distinguish primary types of entities for our analysis: protein chains referring and other non-peptide compounds (including nucleotide polymers like DNA and RNA) as ligands. With this distinction we classify an edge in a hRIN depending on the entities the nodes the edges connect span. An edge that connects to residues within the same protein chain is referred to as an *intra-chain* edge or interaction. Should an edge connect nodes in different protein chains, it is said to *inter-chain*. Finally, an edge connecting a protein chain to a ligand are referred to as *ligand* interactions. For the purposes of our analysis we only consider intra-chain contacts to be interactions within chains in the set of homolog proteins, not contacts within chains said proteins may be in contact with. The reasons for this are discussed later in this section. Furthermore, interactions between non-homolog chains and ligands, and are not included in constructed hRINs.

Breifly mention the EIDN system.

When observing the structures within the PDB for a set of homologs it is common to see closely related chains in complex with varying types and quantities of entities across many examples. Graphically encoding information of all entities into a homolgy-aware network is challenging. There is generally no guarantee that entities present complexes containing a homolog chain in the PDB can be neatly mapped into families across all structures. Given this variability, attempting to encode information about all residues in all present protein in a family of complexes can result in a network that is quite complicated and fails to provide a useful characterization of the family.

To address this issue we implement a system of identifying corresponding entities across structures returned by the homology search - both protein chains and Ligand compounds. These entities are identified in all structures returned by the homology search.

With this in mind we can provide a mathematically rigorous definition of the hRIN object that proves useful in later analysis. In practice, there is much information contained within the nodes of an hRIN, including: 3D position information, consensus residue, residue type.

5 Analysis Tools

5.1 User Defined Queries

Given the wide and diverse set of analyses once could potential want to perform on interactions within a protein family, HomologyRing supports users providing a custom query, allowing for the creation and analysis of a sub graph of the initial hRIN created by the pipeline. This sub-hRIN is defined by 3 primary parameters: `interaction type`, `interaction class`, and `region`. Filters on `interaction type` refers to the specific kind of interaction reported by RING. These are listed and described in Appendix B. This allows users to create a subgraph from a subset of edges of given types. The different types of non-covalent interactions see different distributions within the PDB. Much of the disparity can be attributed to some interactions like `IONIC` and `PIPISTACK` may only occur between a specific subset of amino acid residues, but the different interactions are seen to be associated with different structural and functional features. RINs constructed from all possible interaction types tend to be quite dense, primarily as a consequence of prolific nature of interactions like `VDW` and `HBOND`. The former Van Der Walls interactions are generally present when two neutral compounds are into close proximity and can be either attractive or repulsive. The ladder hydrogen bonds are most commonly responsible for secondary strcture elements within a strcture such as the formation of alpha-helicies or beta-sheets. These tend to be generally well perseved within a family of proteins, but can exhibit notabale conservation and variation at binding sites where the homolog chains interface with either another protein chain or a ligand. A protein family that contais many homologs or paralogs that come in complex with a verity of different entities may have a tendency to exhibit interactions spesific to a given entity. where protein chains that are more consistently observed in the same complex will have these sites conserved. We later show this to be the case in an example where a famaly of human DDX paralogs are observed to have conserved intra-chain `HBOND` interactions at binding sites critical for protein function. Lastly, Hydrogen bonding is also seen to stabalize tertiary and quatary structure to a less common extent. These interact tend to be more variable in families with greater sequence dissimularity. Some of the more novel interactions such as `SSBOND` are seen to have significant conservation within a family.

move discussion of this to the RING appendix. Be breif here

Specification of `interaction class` allows the user to construct a graph out of a subset of edges conditioned on the entities the edges are between. Valid choices are intra-chain, inter-chain, chain-ligand contacts, or all and select the respective edges described in Subsection 4.2. Intra-chain contacts include interactions and corresponding edges between residues in the same protein chain. Consideration of exclusively these edges enable analysis of secondary and tertiary structure variance and conservation within a family. Inter-chain contacts refer exclusively interactions with participating residues in different protein chains. At a glance, conservation of contacts can deliver insight into which contacts are key to the binding of protein chains to form ligands. This enables an analysis of binding domains to gain insight into which residues are evolutionary important for interactions or what adaptations in specific sequences may contribute to specificity in partner binding.

Ligands Metal-coordination can be essential for structure and contain important functional information about a family of proteins. Co-factors DNA/RNA

Finally users may place a filter on nodes to be included in the resulting sub-hRIN. The default behavior is to perform not filtering and include all nodes. Regions of interest in a family may be specified by the user to create a sub-hRIN that captures interactions with a set of residue nodes. For example, creating a hRIN for an entire structure may provide a lot irrelevant information when performing an analysis on an inter-chain binding substrate conservation within a family. Can specify an occupancy ratio, selecting residue-nodes in the network that correspond to MSA columns that contain above a given threshold of non-gaps. This allows users to quickly discard nodes corresponding to the poorly conserved tails of the MSA that are not representative of the family. It should be noted that region filters placed on the nodes are not strict, and should an edge have at least one node in the selected region, the edge and both nodes the edge connects will be included in the resulting hRIN.

5.2 Contat Mask Plots

A very quick means of identifying groups of conserved interactions within a family of proteins is the use of *contact mask plots*. These display an image of the MSA used to define the residue nodes in the hRIN. Intuition about the characteristics of each family may be gained by the use of CLUSTAL colors which encodes physio-chemical properties of different residues into color. On top of this is a binary mask indicating that the residue in a given structure and residue family participate in an interaction that matches the users query.

5.3 Contact Conservation Plots

To quantify the conservation of an interaction within a protein family, Figure 3 graphically shows the conservation score as calculated in Section [6]

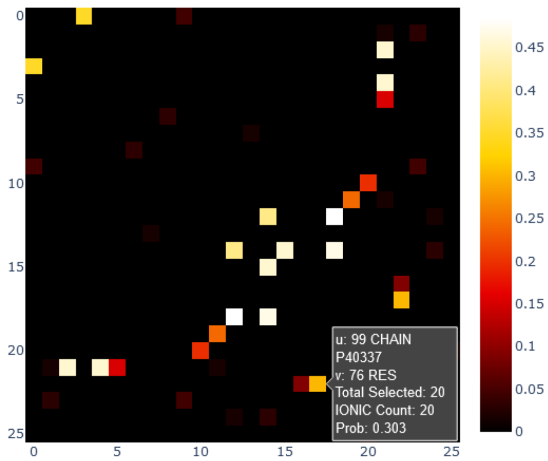

Figure 3: Example of *contact conservation plot* showing the scores for `IONIC` and `PIPISTACK` interactions for all node types: `RES`, `CHAIN`, and `LIG`.

5.4 Co-Evolution of Contacts

With the importance of non-covalent interactions for conformal and functional properties of a protein, a reasonable analysis task is to analyze the conservation and specificity of interactions witin a family by observing the presence of interactions within spesific homolog structures. To achieve this, we provide a means means of classifying structures into a pseudo-taxonomy, grouping structures based on homology-aware RIN similarity. Typically, evolutionary relationships can be inferred by sequence similarity: using disparities between multiple residue sequences to infer the elapsed time since a common ancestor. Further structure is given to these relationships by organizing individuals into a tree to map their relationships. We utilize this concept to compare and group structures based not on sequence similarity, but rather the similarity of their respective RINs.

At the core of this approach is a means of quantifying the dissimilarity of RINs based on their graph definition and the node maps integral to hRINs. Given a hRIN $G = (N, E)$, for homolog protein structure x we define $E_x \subseteq E$ to be the set of edges observed in structure x. We continue to define the edge difference set, $D_{x,y} \subseteq E$ for two homolog structures x and y to be the set of edges present in either x or y exclusively. This can otherwise be thought of as the symmetric difference of the subgraph edgesets:

$$D_{x,y} = (E_x \backslash E_y) \cup (E_y \backslash E_x) = E_x \oplus E_y$$

Visulization of this operator is shown in Figure 7. Observe, edges are considered to be shared between two graphs if their incident nodes have the same labels. We are able adapt this notion thanks to the node maps created with the MSA, and may avoid using more complex and computationally expensive notions of graph similarity. Additionally, the result may have node set of the result will be the union of the two graphs.

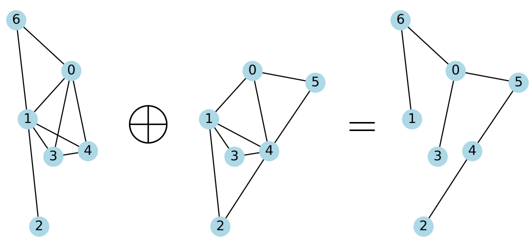

Figure 4: Example of the graph *Symmetric Difference* (XOR) operator on graphs.

With this we define both a unweighted and weighted pseudo-distance[2] to quantify the dissimilarity of homolog chains based on the interactions the admit.

$$d(x, y) = |D_{x,y}| \quad \text{or} \quad d(x, y) = \sum_{e \in D_{x,y}} p(e)$$

[2]This cannot be called a true distance metric, as the triangle inequality does not neccessarially hold. However, does not represent a significant issue for practical issue.

This is also referred to as Hamming distance, though it is worth noting that the weighted version in this context differes from most uses of the Weighted Hamming Distance, where the difference in edge wieghts between the two different graphs are summed. Here the edges, if they are shared between the two subgraphs, will have the same weights, and will not contribute to the distance.

The ladder weighted variant will penalize more discrepancies of highly conserved edges - considering homologs with edges , where the unweighted variant considers only the number of unshared edges significant. The type of analysis being performed should inform the choice between the two. Studies of contact conservation will desire that structures that admit highly conserved edges will be grouped closer together in the resulting taxonomy, while studies of specificity will value shared novel or uncommon edges more important when grouping structures.

With the distance metric defined, the sequences organized into a rooted tree, or deprogram, using the popular unweighted pair group method with arithmetic mean (UPGMA) method. This is an iterative method, employed here to create a heigharchial clustering of structures based on non-covalent interactions of interest. At each iteration, the method defines a distance between groups of examples based on the average inter-group distance:

$$d(X, Y) = \frac{1}{|X||Y|} \sum_{x \in X} \sum_{y \in Y} d(x, y)$$

This is referred to a pseudo-taxonomy because it does not aim to capture inferred evolutionary relationships, rather group structures based on the similarity of the non-covalent interactions the admit. This is useful for finding patters of contact conservation or specificity. For example: conformal variations of a protein complex may give rise groups of variably conserved interactions.

replace fig with example dendro-gram

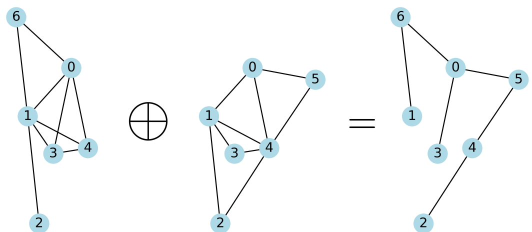

Figure 5: Example of dentrogram created to created structural similarity based on residue interactions.

6 Example Analysis

6.1 PAS Domain - Binding Specificity

6.2 Elongin C - VHL Interaction

In this section we demonstrate the utility of HomologyRing to quickly give structral insight to particular protein family and the complexes it forms. These are observed primarially through the study of inter-chain interactions, and can deliver insight into the spesific sites contributing to protein-protein interactions. We will use the interactions formed by the RNA Transcription Factor B complex, to demonstrate these capabilities. Transcription Factor B, also known as SIII or simply Elongin, is a protein complex responsible for promoting RNA polymerase II elogation by suppressing the function of arresting sites within a transcription window [?]. In this sense it has been identified as an elongation factor regulating transcription by RNA polymerase II. This transcription process is most commonly thought of as being regulated by initiation factors. However, regulation of the process during elongations is increasingly understood as being significantly important. The Elongin complex is seen to be dynamic and known to have several other interacting factors affecting elongation behaivor. We will primarially focus our discussion on ELOC. Using it as a query for a HomologyRing search inorder to observe inter-chain interactions in the Transcription Factor B (SIII) and related complexs. Comparing structural insites gained through HomologyRing to results from the literature will be used to demonstate the usefulness of the tool and information contained in the resulting hRINs.

It this case, it is worth noting that we are not primarially concerned with the construction of a family of homolog proteins per se. Rather, we can expect a BLAST search of the PDB to return many ELOC orthologs, or human isoforms (of which there are three [?]), resulting in a family of high sequence similarity, but may very considerably in the entities present in each complex structure.

The SIII compex is composed of three elongins: A, B, and C and are highly conserved in Eukaryotes. These elongin protein products are respectively refered to by their short gene names: ELOA, ELOB, and ELOC. Behaivor of the SIII elongation factor is well studied experimentally, and has enjoyed a lot of attention. However, much of its function on a mechanistic level is yet to be fully understood. ELOA is the is responible for binding in the SIII Elongin complex with the larger RNA Poly II complex where Elongins B and C are regulatory subunits.

ELOA as a monomer has been seen to promote elongation by RNA Pol II alone

ELOB and ELOC are seen to form a subcomplex [?]. Ubiquitin-like protein SCF ubiquitin ligase subunit Skp1 Seen to promote the activity of ELOA

Figure 6 shows an ELOB - ELOC complex interacting with a portion of pVHL.

We show a complex of ELOB, ELOC, pVHL incomplex with a region of HIF in Figure 6.

Notabale regulation activity comes from the SIII interaction with the Von Hippoel-Lindau Tumor Suppressor Protein (pVHL) which is coded for by the VHL gene. Elongins

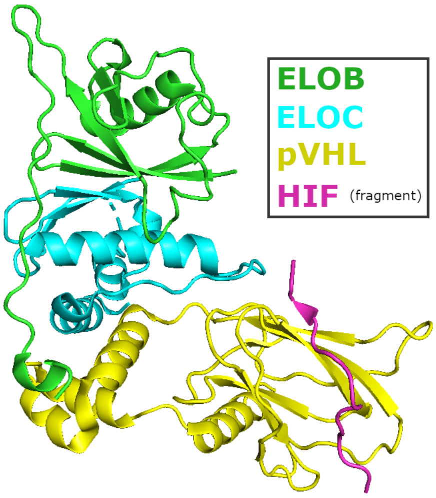

Figure 6: PDB: 1LM8; Complex of Elongin B,C with pVHL and HIF-region [5].

B and C have are known to interact with pVHL in the SIII complex. pVHL, in this interaction is seen to inhibit transcription elongation. The pVHL product is well-known a regulator of many different pathways and known to interact with several other proteins [8]. However, the protein has recieved much attention for its role as a tumor suppressor [**?**], and owes its name for its associated Von Hippel Lindau syndrome – an inheritable genetic mutation to the VHL gene which is associated with a very high propensity for tumor devlopment in many highly vascularized tissues throughout the body.

At the protein level, VHL is known for its role in regulating tissue response to hypoxic, or low oxygen, conditons. Spesifically, pVHL is seen to be contribute the the ubiquitination of hypoxia-inducible factors (HIFs). HIFs are transcription factors, or proteins that bind to spesific regions of DNA to regulate their transcription. They are a key component the process through wich mamilian cells respond to hypoxic conditons, or prolonged leveles of low oxygen. Hypoxia is highly relevant to many forms of cancer, as tumos growing and eventually cutting themselvs off from blood supply in their interior are a common pattern. HIF, working as a transcription factors, can promote the thranscription of various different genes which may serve to mitigate the effects of hypoxia, correct it, or signal for the destruction of the cell. One known mechanism for addressing the issue of hypoxia in tissue is signialing for the additional vasculature, thus increasing the supply of blood and oxygen to the area. This behavior is highly characteristic of the tumors seen in VHL syndrome, occuring in highly vascularized areas such as parts of the brain, retina, and spine, among others.

Healty pVHL, in complex with ELOB and ELOC, tags HIFs with Ubiquitin, signiling for their destrction, and thus working to down regulate a tissues hypoxia response. Mutations to the protein such as the ones seen in VHL syndrome

HIF contains a highly conserved proline containg region which is the binding substrate where the HIF will bind to pVHL. In hypoxic enviroments, the prolines in this region will experice hydroxulation, a post translational modification where proline will gain an hydroxyl groub on the sidechains gamma carbon. This modification enables pVHL-HIF interactions and subsequent ubiquinition of HIF, a process that eventually leads to the destruction of the HIF, freeing the gene region to which it was previously bound. This process is critical for allowing cells to variably express genes based on the amound of oxygen avaible to the cell, allowing for behaivoral changes that allow for its survival like angiogenisis or alternatively programmed cell death in low oxygen senerios [2].

This well-studied complex has recieved much attention for its role in several cancers. It is important to note that a homology search of the PDB does not neccessarily return a set of distinct, evolutionarily related sequences. Instead, the Homology search as implemented by BLAST reutrns sequences with high sequnce similarity to the query. This will most likely many different structures for the same protein, as the PDB may contain many structures for the same protein. This is desiarable behaivor, given that tese different enteries may correspond to the protein in different biological contexts, with differences in temprature or pH; additionally a protein of interests may form different complexes with many interactors, or experience conditional conformal variations. All of these factors influcence the structure and corresponding RIN of a protein. Given this networks with noteworthey differences in topology can be seen to represent the same protein chain. By coreating a hRIN for a sigle protein, we obtain a single object that may be used to represent a protein that captures both variations and conserved features of a protein.

The preceding sub-sections serve to illustrate the many complicated interactions in which ELOC participates. Each of these interactions is quite nuanced, and a complete picture of all possible interactions and their functioning at a molecular level is a very time consuming process to come to understand. Furthermore, in this case there is still much that remains not compleatly understood. Here we will utilize HomologyRing to quickly an object capturing structural information about protein-protein interactions at the residue level. This may be used as an overview, providing researchers a quick introduction to wich sites and compounds are involved in interactions in a protein family, or as a technical, well defined mathematical object, wich could be used for more quantitive analysis. In this section we will demonstrate the former, using homologyRing to quickly gain initial insight in to the Elongin Complex.

Since we are here concerned with interactions of multiple protein chains, we are required to restrict the use of the pipeline in this case to a homology search of the PDB, as predictions avaible in the AlphaFold Database are limited to single chain structures, preventing the study of inter-chain interactions. Bias in the PDB in some ways limit the usefullness of the quantitive information in the graph. The probability weights in the inter-chain edges of the graph are heavily influenced by the frequency with which published experimental structures have captured any given two entities in a single structure. As discussed, in the Elongin complex, this is highly variable Presense of particular entities in experimental structures is also influenced by factors such as interest in particular aspects of a complex within the research community and difficulty of chrystialization. Both of these factors distance the edge weight data from being effective measures of the propensity for an entities interaction in a biological context and such biases should be considered for more statistically rigorous analysis aimed at studying such interactions.

We begin our analysis of the Elongin complex by centering our focus on the ELOC chain, using it as our query structure. And with this we may begin spesifying parameters to construct a HomologyRing quert. Spesifically, the query chain is taken from the `AUTH C` chain of the PDB structure `1LM8`, which is a x-ray chrystal structure of the ELOB-ELOC dimer in complex with a fragment of pVHL [5]. The corresponding query sequence is 88 residues long, and maps to uniprot accention Q15369-2 isoform, differing from the cannonical sequence by an omission of 16 residues from the N-terminal, or beginning of the sequence. The PDB Europe Knowledge Base lists 164 structures in the PDB mapped to the ELOC Uniprot gene: `Q15369`. However, many more close homologs exist with high sequence similarity. Creating an hRIN win an aim to create a comprehensive account of all ELOC structures in the PDB is perfectly feasible with the RingHomology pipeline and can capture infromation from all 164 structures comprising more than 500 MB of information in a few minutes on consumer hardware. However, for the sake of simplicity of visuilization we limit the homolgy search to just 64 structures here.

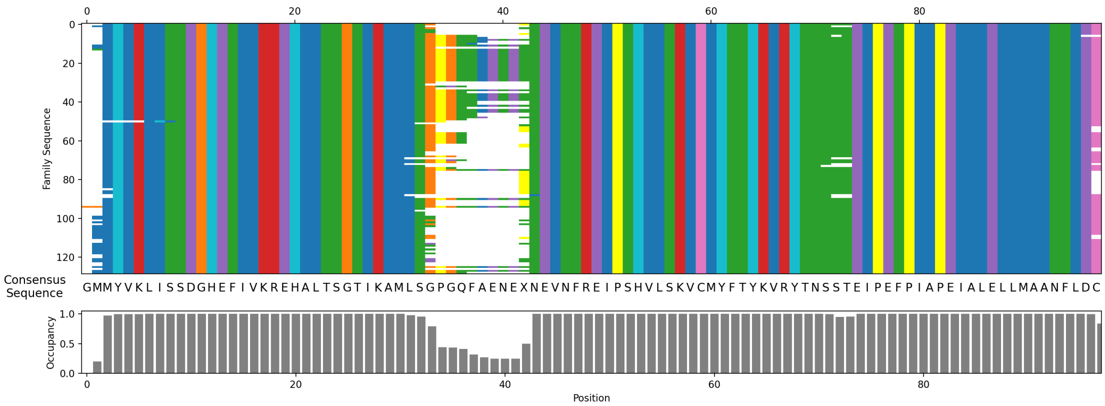

Figure 7: Representation of a multiple sequence alignment for the ELOC family. Created as part of the HomologyRing pipeline with ClustalΩ

Name	**Age**	**Country**
Alice	24	USA
Bob	30	Canada
Catherine	28	UK

Table 1: Common protein chains with corresponding attributes.

We observe the diversity of entities present in structures in our considered hRIN in Figure **??**.

By grouping nodes by their parent protein chain, we obtain a graph akin to a basic protein-protein interaction network (PPI network).

The binding of pVHL to negatively regulates the A conserved region on elongin C consisting of residues 17-50 is needed for the binding of Elongin C to pVHL [7].

Elongins are a member of the ... superfamily Elongin A is primarily responsible for nucleotide interaction and transcription function where the other sub units serve regulatory functions. VHL owes its name with its association with a long known form of cancer inherited cancer - von Hippel-Lindau disease which has come to be associated with with the VHL gene. VHL functions as a tumor suppressor and has many known interactions, and though VHL is not known to have close homologs, it interacts with other genes important for RNA transcription processes [VHLdb][sci-direct]. Among it's known interactions is with Elongin C, which participates in in humans is part of the Transcription factor B/SIII complex. We investigate the specificity of interactions between Elongin C and VHL by initiating the HomologyRing pipeline using the Elongin C present in PDB:

6.3 DDX

DEAD-box RNA helicases, or here otherwise referred to as members of the DDX family of proteins, are critical for many processes involving RNA metabolism and are the subject of much research due to their role in the formation of many cancers. They are responsible for ATP-dependent RNA binding and remodeling, with members assuming general and more specialized roles in RNA metabolism including: RNA duplex unwinding, We examine hRIN created from a set of human DDX paralogues, to examine key interactions between residues in the family, and compare the results to current literature to illustrate the effectiveness of HomologyRing to identify residues and interactions critically important for the function of a family of proteins. In this case, illustrating its usefulness in studying a verity of studies, including virology, genome protection, and cancer. (big sell)

There are DDX homologs are found in all kingdoms of life including viruses; many organisms having many paralogs. Humans are currently known to have 38 (cite) DDX proteins are members of the helicase superfamily 2 (SF2). Intrensically disordered N- and C- terminal domains exhibit high sequence variability within the family, which may contain sequences responsible for regulation, protein localization, or even additional folded domains. Consequently, the terminal regions will exhibit low conservation of sequence and non-covalent interactions.

The DDX protein will bind to the phosphate backbone of one strand of dimerized RNA. The two recA domains of the DDX proteins contain the substrates that interface with the phosphate backbone of an RNA strand. This substrate is variably conserved, allowing for some members of the DDX family to be more generalists, while others exhibit RNA-sequence specificity (super need a citation for that.) ATP binds to a an eponymous D-E-A-D box region of the DDX protein. This refers to a highly conserved Asp-Glu-Ala-Asp motif responsible for ATP binding, mutations of which are known to be highly oncogenic (citation). When bound to an RNA duplex, energy from ATP in complex can be used to introduce a 'kink' in the backbone of one strand in the RNA duplex. The conformal change of the strand induces a local separation of the RNA strands.

In this section, we aim to examine the conservation of *intra-chain* contacts with a family of 37 paralogues and compare conserved interactions with notable conserved structural features characteristic of the family from literature.

In this section, a hRIN is created from a set of user supplied homologs, which HomologyRing supports. This is in contrast to providing a query and letting HomologyRing do a homology search to select homolog structures from a database.

To begin creating the custom DDX family, a query was executed agains the UniprotKB database for entries containing the prosite-PS51192 domain wich corresponds to Helicase 1 ATP Binding Domain. Prosite is a curated database of protein families, domains and, functional sites, wich is useful for getting entries for retreiving structures wit a common feature. In this case, we restrict the Uniprot query to results with structures in the PDB and belonging to Homo sapiens. The results were downloaded and furter filtered in python; keeping only resyts that are annotated in UniProt as being transcribed from a DDX gene. This step is necessary as there are a handful of other genes whos gene products exhibit this domain. The resulting collection of Uniprot entries represents 36 human DDX paralogues with 240 corresponding structures in the PDB. These results are further filtered by using pcbecif's CIF parcer python package to create the final protein family, the subject of our analysis by considering the PDB metadata included in the structure files.

So far we have focused on the homolog use case, which is useful for examining the properties of closely related proteins as measured by sequence similarity. HomologyRing allows the user to provide their own family for which to create an hRIN and perform analysis by giving a list of structures. We display this functionality on a family of RNA dependent DEAD-box ATPases (DDXs) paralogs in Humans. The analysis of paralogs is The family of paralogs has notably lower sequence identity when comapred to common families returned by BLAST homology search with typical parameters. This family has receives much attention for their essential roll in RNA processing

DDXs are responsible for many different roles involving RNA processing. Primary of which is the unwinding of RNA duplexes in an ATP-dependent mannor. This is crutial in RNA transcriptionm splicing, translation initiation, transport, and degradation. As such, they are observed in virtually all living organisms. Eukaryotes will often will possess many paralogs of DDX with many becoming highly specillized for spesific process, while others remain serving more general roles in RNA metabolism.

DDX proteins are also called DEAD-box proteins for a higly conserved Asp-Glu-Ala-Asp motif at the RNA substrate interface. They are responsible for binding and 'kinking' RNA in an ATP dependent manner. The resulting RNA-peptide complex can either part of an RNA binding portion of a larger protein complex, or be resposible for the local unwinding of double stranded RNA duplexes. These sites are curutial for the core responsiblies of all members of the DEAD-box family. There are a handful of other conserved motifs including: ... Additionally the globular core of the DDX proteins are part of the (What is the name of the fold?) and is well conserved. In contrast, the N- and C- terminal regions of members are highly variable and spesific to the specialized functions of members.

Some members are generalists , involved in many or critical proliferic RNA matabolisim pathways . Others adopt a more specialaized role. such as...

The roll of DDX proteins as RNA remodelsers make them cridical for virtually all forms of life. However, they have additionally attacted additional attention for their role in cancer [1].

A great deal of variation is seen within the family to support the specilization of different proteins, but several conserved motifs and domains remain characteristic of the family and well conserved. They all share a common helicase domain that constitutes a globular core.

There are 38 known DDX paralogs in the human proteome [6]. Being paralogues, they are the result of a *gene duplication event*, and have been subject many mutations since allowing for variations in structure and function. The details of this process are provided earlier in **??** DDX proteins are crutial for virtually all processes involving RNA metabolism [6]. . Better word than matabolism?

In this section, we analize the set of human DDX paralogs to identify A set of 38 PDB structures representing DDX paralogues where manually curated. Structures where chosen that have good biological assemblies, meaning most structures where complexes of multiple chains. To initiate the pipeline on a user-defined family, homolog chains to be considered for the analyis must be provided to the HomologyRing tool, and was collected using gene metadata within the collected CIF files. The construction of the hRIN via the HomologyRing pipeline works much the same, except the homology search may be skipped given the provided information.

Observe that highly conserved interactions within the family correspond to regions of critical functional importance.

We find that the interactions with the highest conservation score in the family is hydrogen bonds maintaining a local conformation around the DEAD box - which is critical for protein-nucleotide interface and the functioning of all DDX proteins

7 Usage

7.1 Python

Users accustom to carrying out their work in python may find it useful to work with a hRIN, or associated objects as it is implemented internally by HomologyRing directly in python. To accomidate this, the tool exposes objects to be used by the user, including tabular information in numerous Pandas dataframes **??** or the networkX MultiGraph representation of the hRIN **??**. Historically contact plots have proven useful for for use in algormith and ML contexts. AlphaFold did this... Our parallel notion of *contact conservation plots* promise similar utility. To accomodate this we provide access to this information in the form of SciPy sparse matricies. Optionally storing contact conservation associated with an hRIN in a sparse *dictionary of keys* (DOK) matricies, provides considerable memory efficiency improvements without sacraficing usablility. We this method of tracking edges in a sparce manner is useful even in visualizing them. When creating said plots we optionally allow the user to plot information for nodes which have contacts according to their sparse representation. This greatly aids usability, as a hRIN may at times have more than 1000 nodes, however, a small fraction of which may actually participate in contacts, the resut, if information of non-interaction nodes is not filtered, is a plot where relevant information is difficult to read.

7.2 Command Line Interface (CLI)

Creation of hRINs, either from a user-defined family or by homology search given a query, may be executed via the Python CLI. Following is an example usage for creating an hRIN with `homologySearch` and the minimal required arguments:

```
python3 homologyRing -q <PATH_TO_CIF> -c A
```

- `-q` : Specifies the query CIF file.
- `-c` : Specifies the query chain identifier (default: `label_asym_id`).

To create an hRIN from a user-defined family, the `-F` argument is required to specify either the path of the family TSV file or a directory containing CIF files to be included in the family:

```
python3 homologyRing -q <PATH_TO_CIF> -c A -F <PATH_TO_FAMILY_TSV/DIR>
```

This works well for integration of the output network files with other applications as discuessed in the following Subsection 7.3. Detailed doumentation of arguments is accecible by invoking the help argument python3 homologyRing –help

7.3 File Output

Tabular output.
HomologyRing supports outputing a hRIN – either from an entire network or a user query – to files which can be directly imported into other applications popular in bioinformatics workflows. Cytoscape is frequently used for the visulization of network information arising from biological contexts, including: protein-protein interaction networks (PPI), metabolic pathways, and more recently Residue Interaction Networks. HomologyRing allows exporting hRIN as a pair of tsv files wich allows them to be visualized and analyzed in cytoscape amoung other software. The hRIN is defined in this context with a node file and an edge file, both of whcich contain identifying and metadata information about the entity. In cytoscape, the user may synthize this information with data from other sources, or perform basic analysis on the network.

7.4 Interactive Application

We have aimed to illustrate thus far how HomologyRing as a tool can be used to gain useful information in a wide range of applications. We do acknowledge, however, that knowledge of python and learning the functionality of the methods required to preform analysis represents significance obsticales in its use. To address this, we have an interactive web application that allows a user to access the core functionality of the software in an rich interactive manner.

The web app is ran locally from the users machine in implemented in plotlys Dash — a powerful python framework to build data driven applications in a reactive paradigm where both the front and back end are handeled in Python.

The use of this software allows useres to explore interactions in a protein family in a dynamic and intuitive manner. As a user builds a query spesifying the interaction types, classes, and regions to be included in the hRIN – The supplamenting visulizations are updated. These visualizations include: A 3D display of the netowrk, a contact conservation score plot, and the contact plot. One of the primary appeals of Plotly is the ability to enrich visualizations with additional data such as hover-text to plots. Here, this adis use usability of contact plots and conservation plots, as hovering over a cell, provides the user with information pretaining to a perticular, node, edge, or record corresponding to a structure. To illustrate this usage, Furthermore, there is the addition of click events, wich are utilized by the interactive HomologyRing app to allow the user to create dynamic selections. Should the user click on a cell in the *contact conservation plot*, an set of multi edges associated with that cell become selected; and clicking a cell in the 3D network plot results in a node being selected. A summary of information pertaining to the users current selection is displayed to the right of the 3D Network isualization. Summary content varies depending on the type of object selected. Should a edge be selected, breif information identifying the nodes that edges are between are provided. Additionally interaction conservation score is displayed and information of interaction observations in structures caputred by the edge is displayed in a tabular format below. Observations are grouped by interaction type, and the user has the option of eaither displaying a count of obervations of interactions of that type is provided, or a list of structures said interactions are observed in. Alternatively, should a node be selected, more detailed information about node identifying information is provided. For the different types of nodes this information will vary. All nodes will display their numberic id used to identify the node within the hRIN. For residue type nodes, this id corresponds to the column of the MSA the node represents. Additionally an amino acid residue from the consensus sequence is given, giving the user indication as to the properties of the residues in that family. Nodes representing protein chains, the gene the protein is transcribed from along with the DB asccention – typically UniProt or PDB – is given along side the protein name where the information is available. Finally, names of a selected compounted identified by a `LIGAND` node is displayed.

The 3D visualization allows users to see nodes and how they generaly releate to eachother spacily. Interactions specified in the user query will be displayed in the graph. It is important to note that only nodes connected by an edge returned by the user query will be displayed in the plot. This reduces visual clutter, and also reflects the internal representation of the custom hRIN defined by the user that may contain information of pirticular interest to their use case. The 3D representation of the network takes position information from the Query chain the user specified when the pipeline was initiated. It is often the case that the MSA will open gaps in the query sequence, and as a result, there will be no position information associated with some nodes for use in the visualization. To solve this issue, the positions of nodes that do not have a member residue in the query chain are linearly interpolated or extrapolated from other nodes position information. Finally, the user can choose to display the 'backbone' of the polypeptide chain. this is displayed as a polyline connecting the positions of the alpha carbons in each residue in the chain, allowing the user to see the conformation of the query chain and infer faimily structure when only a sparse subset of nodes may be displayed.

The query builder functionality is divided into two spaces Edge filtering options are placed to the left of the *contact conservation plot* and node filtering options are placed above the *contact mask plot* which provides a visualization of the MSA that defines the residue type nodes. Filters on the edges include allowing to users to spesify the interaction class as in **??**, and the interaction types as discussed in Section **??** are the primary tools for spesifying the interactions of interest to be included in the displayed hRIN. The interaction class dropdown allows users to focus their analysis on all possible interactions to form edges, or limit results to intra-chain, inter-chain, or chain-ligand interactions. Just below is a table containing basic information on the types of interactions present in the protein family. This display the number of interactions of each given type and the ability to toggle inclusion of the interactions in the network.

Above the *contact mask plot* are controls for region filtering. The user may either choose to include all nodes, spesify a range in wich residue nodes will be included, or an occupancy ratio may be given. The ladder most option will trim ends of the MSA and associated nodes from the network where the *occupancy ratio* of the MSA or the fraction of non-gap resodies to total residues that fall below the given ratio. For all node filtering methods, it should be noted that nodes not matched my the node filter will be included in the output network if it is connected to a matched edge by a valid interaction.

8 Formal hRIN Definition

In practice, a RINs and hRINs contain a considerable amount of metadata which is utilized trhoughout the HomologyRing pipeline and supporitng analysis. However, the details of wich are not needed for a functional understanding of the hRIN object. Furthermore, since RINs show great promise in ML contexts, we provide a more formal description of the hRIN object that lends its to future work in these contexts.

We use the variable x to represent a homolog and its corresponding structure information. There is notoriously a great deal of information contained within the CIF file, so a riggerous definition of this object could get quite tedious. Thankfully, here primarilly use the variable to identify homologs within the family for which the hRIN is built, which we call X. For example, a homolog search executed on the PDB will return a family of structures such as $X = \{\texttt{8bb5\_B}, \texttt{8bb4\_O}, \texttt{8je2\_C}, \ldots\}$

We may then formalize our abstraction of the RIN output of RING ran on a single structure as a multigraph:

$$H_x = \{V_x, E_x, \varphi, c\}$$

Where V is the set of nodes in the RIN, E is the multiset of edges, φ is the incidance map, and c is the edge typing function. At this point, nodes in V represent entities as ndoes as identified by RING, which provides a ringEdges output filie providing metadata about the mapping of components in the CIF strucutre to nodes in the RIN. In practice, RING will obviously identify idividual peptide residues in a chain as nodes to form a RIN, but furthermore, RING is notable for its ability to handle incorperate information about multiple protein chains and ligands into the network. The set E The incidance map, $\varphi : E \to V \times V$, is used to relate a given edge to the pair of verticies, or nodes, which it connects. For example: $\varphi(e) = (v_1, v_2)$.

This notation borrows from a common formalism of multi-edge graphs where edges may be colored, where the map c is used to identify the *color* associated with a given edge. Here, $c : E \to T$, is imployed here to map a given edge to the type of interaction it corresponds to. We denote arbitrary interaction types $\xi \in T$ where T is the set of possible interaction types predicted by RING such as: `HBOND`, `IONIC`, `PIPISTACK`.... A complete list of possible interactions that define T are provided with detailed conetxt about their properties is provided in Appendix B.

Importantly, such a multi graph H_x is created for each homolog in the family $x \in X$. We denote the collection of such objects as H which we will often refer to as the family of structures, or simply family. It is worth noting that H is simply a collection multi-graphs, and does not contain any of the homology information at the residue level necessary to create the hRIN. Transforming this information into an hRIN requires a series of maps, which in the implementation of HomologyRing, is handeled by the MSA created by ClustalΩ and the EIDN system.

The EIDN system is a means of identifing non-homolog chain entites present in a family of structures in a consistent manner. It was developed for HomolgyRing in order to map such ligands and non-homolog peptide chins to nodes in hRINs. It functions by compiling a list of all entities present in a family. In CIF files, entities present in structures are classified as either polymer or non-polymer entities and each entity witin a structure is assigned its own numeric EntityID. Polymer entites include protein chains and nuclaic acid chains such as DNA and RNA; where all non-polymer enties are labeled as ligands. Of important note is the fact that RING, and by extension HomologyRing, considers non-peptide polymers such as DNA and RNA as ligands. Once all entities across all structures in a family are collected, the entities are then grouped either by the name of ligand compounds or the gene that non-homolog protein chain was transcribed from. After this group operation, the results are then enumerated — assigning a numerical identifier to each non-homolog entity. To avoid confusion of these numerical identifiers with the system to identify `RES` type nodes and unite the system into a single numerical identification system, we add the length of the MSA to non–residue type nodes.

For example, if a families MSA has 250 columns, and 20 non–homolog chain entities, the resulting hRIN would have 270 nodes. Nodes with id 0–249 will be of type `RES`, and nodes 250–269 will each be of type either `LIG` or `CHAIN` and their ids are assigned by the EIDN system. The node ID is primarially used internally, but exposed to the user for refrence. More practically, the user will be more concerned with either the properties / location of residues families, or other information about compounds identified with the EIDN system. For this reason we provide a map, relating EIDN nodes to the gene a protein chain was transcibed from or the name of a compound a non-residue node represents.

We then combine these to systems to identify residue families as given by the MSA and families of chains and ligands given by the EIDN system to define a map $f_x : V_x \to V$ that for a given structure x, relates nodes in its RIN H_x, to the set of homology-aware nodes in the hRIN which we denote V and take to be the union of nodes identifying residue families given by the MSA and nodes given by EIDN. Alternatively, V can be seen to be the union of the image of all f_x:

$$V = \bigcup_{x \in X} f_x(V_x) \tag{1}$$

Which includes all MSA columns and all non-homolog entities accross all structures in X.

Observe that f_x is a graph-homomorphism — Formally, a mapping between (multi)graphs G_1 and G_2 such that verticies in G_1 are mapped to verticies in G_2 and edges in G_1 are mapped to edges in G_2. This is frequently denoted as function of the verticies of the two graphs, $f : V_1 \to V_2$ such that if $(u, v) \in \varphi(E_1)$, then $(f(u), f(v)) \in \varphi(E_2)$. Additionaly observe that taking the family of such maps, given by the family of structures, admits a family of graph-homomorphisms onto a single graph, namely the hRIN of the family. This family is at the core of the definition of the hRIN for a family of structures:

$$F = \{f_x : V_x \to V \mid x \in H\} \tag{2}$$

Simiular to the definition of V, we use this family of homomorphisms to define the edgeset of the RIN:

$$E = \{(f_x(u), f_x(v)) \mid (u, v) \in E_x, x \in X\} \tag{3}$$

This is then sufficient to give a formal definition of the hRIN object:

$$\mathcal{H} = (V, E, \varphi, \rho, c)$$

Where V is the set of nodes and E is the set of edges as defined in equations (1) and (3) respectively. $\varphi : E \to V \times V$ is the incidence map, which given an edge e reutrns the pair of nodes the edge connects. Finally, $c : E \to$ is the *edge type map* and is invarient under homomorphism, meaning the type is the same as the edges in the pre-image.

Additional notation that proves useful in this paper is the definiton of the $\mathcal{H}_x$. Which can be seen to be the the homology-aware analog of H_x:

$$\mathcal{H}_x = (f_x(V_x), \{(u, v) \mid (u, v) = \varphi(e), e \in E_x\}, \varphi, \rho, c)$$

Where ρ is recalculated on the subset of edges E_x. This can also be seen as the subgraph of $\mathcal{H}$ admitted by a pirticular structure of interest x which will only contain nodes corresponding to non-gapped residues in columns of the MSA or entities present in the structure file for x.

This is used to define a dissimilarity measure in Section **??** to group structures based how simular their interactions are.

Appendix

A Biological Background

A.1 Central Dogma

Amino acids, different properties Proteins transcribed from genes - DNA that codes

A.2 Protein Folding

Proteins are fundamentally polymers, or variable length chains of amino acid units joined together by covalent bonds. The individual amino acids in a chain are referred to as residues. They all have two primary componets: a backbone consisting of a central carbon atom with two essential groups on oppsite sides: an $-NH_2$ amino gorup and a $-COOH$ carboxyl group; and a variable side chain that is principally responsible for the physical and chemical properties of the residue. Proteins are bonded together in a chain

Figure 8: Example of an amino acid structure.

by forming peptide bonds — a perticular kind of covalent bond where the amino group of one residue will bond to the carboxyl group of another. The sequence of the spesific residues in a protein chain is referred to as the *primary structure* of a protein.

There are 20 standard amino acids that primarly differ by their side chains. These side chains have varying polarity, charge, and size which effects the chemical and physical properties of the residue.

Figure 9: The standard amino acid residues, groupped by properties.

Secondary Structure

Secondary structure of a protein refers to the most common local conformal patterns observed in the polypeptide chains constituting proteins. These patterns promote hydrogen bonding between atoms in the back bone of residues creating the beginnings of a structural scaffold. The most common of these patterns are α-helicies and β-strands. α-helicies are right-handed spirals stabalized by hydrogen bonds formed between the oxygen of a carboxyl group and the hydrogen of an amide of another residue.

β-strands are linear chains of typically 5 to 10 amino acids. steatched out alternating side chain orientation

β sheets can either be parallel or anit-parallel. parallel slightly less stable

Figure 10: The standard amino acid residues, groupped by properties.

Tertiary Structure & Protein folding
Tertiary structure refers to the oerall three dimensional conformation a protein adaps as it fols. This folding is dirven by a process influenced by the cellular enviroment and interactions between the different side chains of amino acids in the polypeptide sequence.
Quatrary Structure & and protein complexes
Quatrary structure refers to the interaction between different polypeptide and other non peptide conpounds, or ligands to create protein complexes. These complexes It may be the case that a protein chain may occur in only one complex, where as some proteins may have a large network of interactiors with which it is observed in complex with. We often refer to the spesificity of such interactions to describe the popensity of a protein to bind with only particular partners.

Complexes of multiple proteins are fromed primarially after a protein has folded and are they result of interactions between residues across the different protein chains. To facilitate partner binding, proteins will have binding substrates with motifs, or notable amino acid sub-sequences, to facilitate a particular interaction. These regions or motifs are often critically important to the function of the structure

Protein structure - function relationship The final stucture or conformation of a protein is intrensiclly linked to the function of a protein.

A.3 Homology

It should be understood how several organisims may be evolutionarly related; from one common ancestor, mutuations are accumulated that may affect an individuals fitness, or the ability to thrive for a given enviroment. These mutations are often neurtral, having not have much if any obsevable effect; however, mutations that alter fitness are most likely to be negatvie, but some mutations may confer a slight benifit to the individual and become selected. Accumulation of these mutations in a sub-population can lead to notably different traits and eventually speatiation. The discussion of these evolutionary relationships are well-defined at the orrganism level, but also do much to abstract away the low level on which changes occur. At the molecular level, proteins evolve through small changes that may lead to functional variations or the formation of evolutionarily related protein families. Mutations in the DNA of a gene can alter the protein sequence, while structural mutations like insertions, deletions, and inversions may lead to more significant changes. This process, known as *molecular evolution*, descibes several different ways in which different protiens may be related.

Homology is a general term that simply indicates that two or more proteis are related through a common protein ancestor. The common ancestor may be quite distant, even occuring befror a speciation event resulting to homologs in different species called *orthologs*. These genes and the proteins for which they code often serve similar functions in the different species. *Paralogs* are genes within the same species that arise from duplication events of a sigle ancestral gene and result in, initially, two copies of gene and their protein product within the same species. They may diverge over time to adapt different structure or function, but often retain similar ones. Paralogs allow for diversity of similar proteins within a similar species and permit specilization in function.

Homologous proteins are often observed to functional and structural similarities, as mutations that radically impact the function of a protein are unlikely to confer an advantage. The theory of homology is frequently used to classifiy proteins into groups or faimilies usiving several different methods to form a taxonomy.

Record providing direct evidence for two proteins being homologs most often cannot be established. Instead, homology is typically inferred through similarity between sequences. A common task is to find set or family of sequences related to a given query sequence through a *homology search*. One such tool for performing these searches is BLAST, or Basic Local Alignment Search Tool, which identifies regions of high similarity between the given query and other known sequences. The match is visualize trough a pairwise alignment provided for each pair of query sequence - subject sequence match returned. There may be millions of sequences of a database, and inferring homology through a pairwise alignment can be a very difficult and computationally intensive problem. BLAST utility and prolific use can largely be attributed to its efficient implementation of its search algorithm, allowing for fast and accurate set of homologs using a heuristic search method. Details about different databases available for homology search and their content, as it pertains to this paper, is discussed in Section 3.

A.4 Multiple Sequence Alignments

Multiple Sequence Alignments (MSAs) function to align residues in a set of sequences. These could be sequences of arbitrary symbols. However, MSAs see extensive application aligning sequences of nucleotide sequences such as DNA or RNA, and protein sequences. Nucleotide sequences use alphabets of A, T / U, C, G as stand-ins for nucleotide residues — In DNA, these are adenine, guanine, cytosine, and thymine, respectively. Where in RNA sequences the same residues are observed with the exception of Uricil, denoted with a U, is seen where Thymine is not observed. Amino acids in a protein sequence are identified in this context with a single letter code to designate each residue, as given in Figure 9. There are many different tools for creating MSAs but all generally work by inserting gaps, designated by a '-', into the sequence. Creation of a MSA on a set of sequences typically assumes some form of homology, or evolutionary relationship between sequences in the set. A consequence of this assumption is that evolutionarily related sequences will exhibit some degree of similarity. The use of MSAs allows for examine conserved sites or motifs in a family of sequences. When dealing with protein sequences, these conserved features often correspond to structurally or functionally important aspects of a protein.

A.5 Structure Files

The ability to resolve the structure of biomolucules has been transformative to biology as a whole—giving invaluable insigths to the functioning of processes on molecular level. Structural information is now a core aspect in a wide range of fields, providing a deeper understanding of molecular mechanisms that underpin life, from cellular processes to the development of novel therapeutics in areas such as medicine, biochemistry, and biotechnology.

Protein strctures are determined experimentally using methods such as X-ray chrystallography, nuclear magnetic resinance (NMR) spectroscopy, or cryo-electron microscopy (cryo-EM). X-ray chrystallography requires a purifed protein sample to be chrystallized. Passing a high-power X-ray beam through the chrystal results in the creation of a diffraction pattern that is used to determine the 3-D structure of a protein through numerical methods like inverse Fourier Transform. This results in high quality position information, with enough spacial resolution to accurately determine the position of individual moldecules. The need for a high purity chrystal sample, however, can pose a significant limitation, as proteins with dynamic conformation such as intrensically disordered proteins (IDP), or highly hydrophobic proteins like trans-membrane proteins (TMP) can be difficult to chrystallize, resulting in bias in the structural information avaible using this method [**?**]. NMR explotis the magnetic properties of atoms nuclei to resolve their position. NMR works by applying a magnetic field to a protein sample and detecting the energy released as the atomic nuclei return to their original states, which helps determine the distances between atoms. NMR has the benifit of being able to study proteins in solution, allowing for observation in conditions that may more similarly resemble their normal biological conditions. Furthermore, the technique is particularly useful for observing the conformal dynamics of flexiable proteins. Cryo-EM involves rapidly freezing protein samples in a thin layer of ice to preserve their native structure, then using electron microscopy to capture high-resolution images of the sample. Thousands of these 2D images are collected and combined using computational techniques to reconstruct a 3D model of the protein. This method is particularly effective for studying large molecular complexes and proteins that are difficult to crystallize.

An image could be inserted here if filler is needed.

Accessiability to strucutural information obrained by such experiments has been revolutionary for the feild of Bioinformatics. This is facilitated by standardized file formats for structural data and databases to catalogue and facilitate access to information. The Protein Databank (PDB) is the foremost database for structural data, offering a comprehensive resource for published experimental structures. The avaibility of experimental structural data enabled the training of models such as RosettaFold and Alphafold, which are both machine learning models capable of creating high accuracy predictions of protein structure given an amino acid sequence. Predictions for AlphaFold2 are catalogued in a database and publically avaible. More information about AlphaFold and its structure database are avaible in.

link ref

As previously mentioned, strucute files availible in such databases are stabdarduzed to aid in machine and human readability. One of the notable early efforts to do this was with the PDB file format, wich still enjoys prevelence, but is in the process of being displaced by the more recent PDBx/mmCIF format (eXtended PDB/Macromolecular Crystallographic Information File). In this work we focus on the mmCIF file format, commonly refered to as just CIF, as HomologyRing is made to work exclusively with this file format. The CIF format uses dictionary-based schema allowing for notable extensibility, allowing for effective storage of many different data types. A great deal of metadata are included in these structure files; giving biological context to the structure by giving experimental and author information, indicating what genes and organisim a protein chain was transcribed from, pH information and other physical information, what experimental methods were used, mutation information, amoung others.

Perhaps the most important section of a CIF details the position and identity of atoms in the structure. This is detailed, internally, in the `_atom_site` category, and adhears to a heigharical structure wich is important for the both the biological asseblies of protein complexes and working with said structures in a structural Bionformatics context. In a CIF protein structure, individual atoms will have a set of identifing and associated information like a numerical identifier, symbol corresponding the atoms element, 3-D coordinates, compositon ID, a sequence ID, and chain identifiers. The composition ID, sequence ID, and chain identiefiers are both used to assign atom membership to groups within the CIF data higharchy. This higharchy formalizes biologically relevant concepts used to describe polymers like DNA and proteins into a well-defined data structure with several levels: atom, residue, entity, chian (molecule), and structure (complex). Atoms belong to residues in polymers. In proteins these are the individual peptides which are identified within the `_atom_site` section of a CIF by their composition id with their abbreviations like 'ALA' for Alinine, 'CYS' for Cystine, or 'PRO' for Proline, for example, with a similar system for the nucleic acids. Residues are assigned a numerical index, or sequence ID, with in their polymer chain

Additionally, some atoms not belong to the a macromolecule – polymers like RNA, DNA, or proteins which are most frequently the primary subjects of study in CIFs. such atoms are reffered to as 'hetroganous atoms' or `HETATM` for short. These capture metal ions, water molecules, or ligands, that are biologically relevant to and captured along side the experimental structure. Ligands are molecules that bind to a larger molecule to form a complex and play essential roles in biological processes.

CIFs are supported by a host of software common for structural analysis. Structural viewers like PyMol and Chimera allow for visual interpretation of a structure which is immediately quite useful for gaining an intuition for the conformation of a protein. However, this is still not without limitations: structuers are often large and quite intricate, requiring the use of cartoon representations of chains to limit the amount of visual information presented to a user. Furthermore, intrepiting functional consequence of strucutres often take a lot of time and biological expertiese. This problem is aided by anotations present in structure databases, as well as many different software tools for performing spesific analysis such structures, something we do as well in this work. HomologyRing supports the use of either PDB or AlphaFold Database for homology search. The choice of which to use will depend on the motivation of their analysis. Differences between these databases and the benifitis of each source is discussed in Section 3 .Each

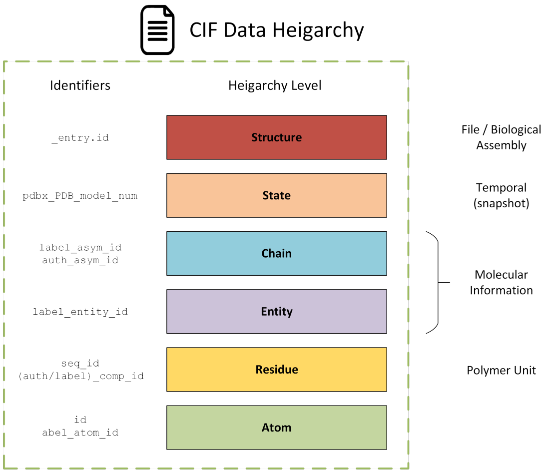

Figure 11: Diagram displaying the various levels of the data organization of structural information in the CIF format.

structure of a family will contain atleast one entity. These entities may be polymers such as DNA, RNA, or polypeptides; or non-polymer entities like water, or other compounds like co-factors. Each entity is given its own numeric internal identifier within a cif file, which are grouped into chains, identified by a character code. Structure files of this form can be quite large, commonly 10s of megabytes, and repositories of structures rapidly become very large. The PDB contains on the order of 300 GB of structural data and the AlphaFold Database is estimated to be between 20–30 TB. This makes performaing analysis at scale of such files, as needed for homology studies potentially quite storage intensive.

The CIF format does has the benefit of being both human readable in text form and machine parceable. However, information about the geometry of the structure is given by the position of discrete atoms. This makes extracting biologically relevant information that one may want from Both of these factors can make inferring relevant information from structures at scale quite difficult. HomolgyRing provides a method for encoding structural information for a family of proteins into a network. The process of which is outlined in Section 4.

B Appendix: RING - Predicted Interactions

RING predicts and classifies various types of molecular interactions based on deterministic factors such as geometric orientation, distance, and the chemical nature of the compounds involved. These interactions differ significantly in their chemical and physical properties, including strength and specificity, which in turn influence the likelihood of different amino acids forming specific bonds. Understanding these variations is essential for interpreting the structural consequences and functional roles of these interactions on the atomic and protein level as well as in the context of RINs and hRINs. This section will define and provide context for all interaction types classified by RING, enabling a better understanding of their role in protein structure and function.

B.1 Van Der Waals Forces - VDW

Van der Waals forces are relatively weak non-covalent interactions between molecules or atoms. In fact, they they are considered one of the weakest interactions observed at the atomic level. VDW forces arise from temporary shifts in electron density induced by the proximity of other molecules or atoms. These shifts occur because of fluctuations in the electron cloud, leading to temporary dipoles. As a result, Van der Waals forces can be either attractive or repulsive. At very small distances, the repulsive forces dominate due to the proximity of electron clouds, while at larger distances, the attractive forces generated by induced dipoles become prominent. This balance creates a stable equilibrium distance, which plays a key role in maintaining weak but widespread interactions.

In proteins, the have prolific presence, as they arise from virtually all residues that would would considered to be in contact. Though individually, a VDW interaction represents a small force, their accumulative effect over large substrates allows them to play a significant role in tertiary and quatrnary structure. They are of notable interpretation in the context of RINs, where a great of protein topology is captured by VDW forces, as they can be used as an indicator of residue proximity. However, in practice, they often prove difficult to use in this way. Working with structure or topology information directly can be a complex task. Furthermore, they provide a great deal of data, with a residue often experiencing VDW interactions with many other residues. This surplus of data often hinders the extraction of useful information. In a protein family, the VDW interactions tend to be poorly conserved, and given their abundance, it is difficult to prescribe functional importance to a subset of VDW interactions in a family. For this reason, we often choose to filter them in our analysis to highlight the effect of more interactions with higher specificity.

B.2 Hydrogen Bonds - HBOND

Hydrogen bonds are most prolific non-covalent interactions second to VDW forces. They are most commonly observed in their role of maintaining secondary structure components: α-helices and β-sheets. Hydrogen bonding is most prevalent in amino acids with polar side chains, where electronegative atoms like oxygen, nitrogen, and sulfur

However, they are also commonly observed among tertiary interactions, stabilizing the overall conformation of a chain, and quaternary interactions to form complexes. Hydrogen bonding is additionally of notable importance for protein-nucleotide interactions [?]. Residues most commonly forming these interactions are serine, threonine, asparagine, and glutamine. [?]

Information of hydrogen bonding in protein families provides is quite useful, despite their prolific nature. it is commonly observed that the preservation of local conformation of functional sites will be critical, even in a family of distantly related proteins. Conservation of hydrogen bonding corresponds to regions of high sequences conservation for interactions of short distance

B.3 Ionic Bonds - IONIC

Ionic bonds occur at the atomic level when an electron is transferred from one atom to another. The resulting atoms, having transferred an electron, become ionized hand have and each have a net charge. The donor will exhibit a positive charge, while the recipient becomes negatively charged. This difference in charge, creats a strong electrostatic attraction, bonding the two atoms. In proteins, ionic bonds are often referred to as *salt bridges*, and their formation generally require the presence of amino acids with charged side chains. These include: lysine, arginine, and histidine with positive charge; and asperic and glutamic acid have negatively charged side chains. This creates a loose relationship between conservation of these residues in a sequence and the conservation of ionic bonds they may participate in in a structure. The ability of the charged amino acids to participate in ionic bonds is pH dependent. This allows some proteins to adapt variable conformations in response to cellular environment; this is one possible functional interpretation of an ionic bond with in a protein.

B.4 $\pi - \pi$ Stacking - PIPISTACK

π-π stacks are spesifc interactions between pairs of aromatic rings; or stable, planar, ring-shaped molecular structures with alternating single and double bonds present in some amino acid side chains. In proteins, such amino acids with requierd for the interaction are: phenylalanine, tyrosine, histidine, and tryptophan. These aromatic rings have decentralized π-electrons, to which the interaction owes its name. These electrons have distributions not centralized around a single atom, as typically the case, but rather the electrons may exist in a shared annular cloud around the aromatic ring. These perculiar electron distributions permit specialized interactions: either with other aromatic rings to form a $\pi - \pi$ interactions, or other compounds, as seen later.

$\pi - \pi$ interactions may have one of two possible configurations of the participating aromatic rings:

- **Face-to-face**: where the rings are stacked directly on top of each other with a slight offset.
- **Edge-to-face**: where the rings are oriented perpendicularly, with the edge of one ring pointing toward the center of the other.

In both cases, the interaction of the π-systems creates an attractive force that stabilizes the residues relative positions.

These interactions tend to be quite specific and relatively well preserved. Often serve to stabilize tertiary and quatrary structure. Given this role they are quite useful for characterizing a families topology.

B.5 π-cation Bond - PICATION

Similar to π-π interactions, π-cation interactions require the particular electron distribution of a π system seen in an aromatic ring. However,

Donor cations can belong to another residue with a positively charged side chain, allowing the interaction to play a notable role in tertiary and quatrary structure stabalization. Beyond intra or inter–chain interactions, π-cation interactions or often important for ligand binding and in enzyme active sites; where aromatic residues can be used to locate cations and stabalize interactions.

both pi cation and pi pi stacking have noteworthy roles in binding with pi systems in ligands

in these interactions the system is to considered to donate a π electron to the cation, which becomes a π-acceptor.

B.6 π-hydrogen Bond - PIHBOND

Similar in action to both π-cation bonds and typical hydrogen bonding, π-hydrogen bonds are stabalizng interactions where a positively charge hydrogen bond donor, such as a polar $-$OH or $-$NH group, interacts with the electron rich π system of an aromatic rings, present if several amino acid side chains. π hydrogen bonds are weaker and far less common than both π-cation and π-π stack interactions. As a result they tend be less frequently important for the structure and function of a protein, and they are observed to often not constitue a significant part or residue interaction networks. However, there are notable exampels where they perform a significant role, such as in DNA binding proteins such as transcription factors and aromatic rich enzyme active sites.

this is interesting. see if you can expand on the DNA binding

B.7 Halogen Bonds - HALOGEN

Halogen bonding is a non-covalent interaction involving a halogen atom (such as fluorine, chlorine, bromine, or iodine) and an electron-rich atom or group (such as oxygen, nitrogen, or sulfur) acting as an electron donor. Although halogens are typically considered electronegative, when they participate in covalent bonds, they exhibit a region of positive electrostatic potential called the σ-hole on the side opposite to the bond. This σ-hole allows the halogen to act as an electron acceptor, facilitating the formation of a halogen bond with an electron donor. In proteins,

B.8 Disulphide Bonds - SSBOND

Disulphide bonds, also called disulphide bridges, are uniqe in this list of interactions as they are covalent bonds; and the amino acid cystine is also unique for its sulpfur containing $-$SH thiol group in its side chain, need for the formation of disulfide bonds. The formation of the sulfur bridge reqires an oxidation reaction where the hydrogen is displaced, and results in a S$-$S interaction that is considerably stronger and more sable than many of the other inter-residue interactions witin a protien. That said, they are reversable under reducing conditions, allowing for dynamic protein conformations that are responsive to certain cellular conditions. The strength of the covalent bond is durable in harsh extracellular conditions, and is often seen stabilizing structure in such enviroments. These bonds are generally seen to have notable separation in a protein chain or present in typically well preserved inter-chain interactions. This gives cystine a clear role in contributing to tertiary and quaternary structure of proteins. Due to the unique nature of cystine and disulfide bridges, cystines are frequently well preserved at the sequence level within a protein family and so too are associated disulfide interactions. Cystine is notable for participating in other interactions such as metal coordination, or enzymatic activity, but where it participates in in disulfide bridges, tend to represent highly conserved sites within a protein family.

cite and more details

B.9 Metal Coordination - `METAL_ION`

Involves the interaction of a metallic cation. Metal Coordination bonds can be covalent, where they are also called dative bonds, or non-covalent. Even non-covalent metal coordination interactions tend to be high strength interactions. Coordinate Covalent bonds are particular because a pair of electrons involved from the bond are donated by a single atom (often nitrogen, oxygen, or sulfur) and paired with an electron-deficient acceptor. Since the donor contributes a pair of electrons, the most common acceptors for this type of interaction are Zn^{2+}, Fe^{2+}, and Cu^{2+}.

Metals in proteins have a key roll in electron transfer processes, allowing proteins to facilitate redox reactions. Finally, centrally coordinated metals often of great strucural significance, as the strong coordination interactions work to stabalise the conformation of a protein as is the case with zinc finger proteins where cystine and histidine bound to a Zn^{2+} ion gives the structure its iconic shape, enabling DNA and RNA binding.

Metal coordination, then, most often represent critical components of a proteins structure or function and consequently are frequently well perserved in families. However, experimental and procedral variation in structure creation can make there presence inconsistent in some senerios. Notably, AlphaFold2 does not predict the presence of metals, or other ligands, so will not be represented in predicted structures.

C Appendix: Installation and Dependencies

The code repository can is available from the BiocomputingUP GitHub:
https://github.com/BioComputingUP/ring-homology-pipeline
Execution of the pipeline has a set of commandline programs be installed and correctly added to `PATH`.

BLAST Command Line	`https://www.ncbi.nlm.nih.gov/books/NBK279690/`
ClustalΩ	`http://www.clustal.org/omega/`
RING standalone	`https://ring.biocomputingup.it/`

Note that ring4.0 command line is needed for use of `use_label_asym_id` argument when building a hRIN. Additionally, commandline BLAST requires copies of databases be built and installed locally inorder to perform a homology search. The two that we display in this paper and are likely to be of particular interest to a user is '`swissprot`' and '`pdb`' which may be downloaded from the BLAST FTP server. Files, typically in FASTA format must be built into blast databases using the BLAST commandline tool and placed into the `blast_db` directory. The user may alternatively create their own blast database with the same method, however, it should be noted that the resulting database may have a different schema and the config file: `config.json` may have to be modified to allow for successful parsing. Specifying that the homology search should be performed remotely with the `remote_BLAST` argument can avoid the need for a local blast installation. However, it should be noted that the search usually takes much longer on NCBI servers.

Trying to deprecate this, if possible

D Acknowledgements

E Future Work

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